RAMYA SRINIVASAN

AI Solutions Architect • Cloud & ML Platform Strategist • Enterprise AI Integration
MS in AI & Machine Learning (In Progress) AZ-900 Certified • AZ-104 & AZ-305 In Progress
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PROFESSIONAL SUMMARY

MS in Artificial Intelligence & Machine Learning candidate (Indiana Wesleyan University, 2026) and seasoned AI Solutions Architect with 10+ years of hands-on experience delivering enterprise AI/ML systems across US Healthcare and Retail E-Commerce. Proven impact includes a 40% reduction in administrative effort through an enterprise AI virtual assistant and a 25% reduction in customer support volume through an LLM-powered conversational AI solution. Recent work spans Azure OpenAI, Retrieval-Augmented Generation (RAG), and GenAI document intelligence. Microsoft Azure Solutions Architect Expert (AZ-305) certification in progress, building on AZ-900 foundation. Deep expertise in end-to-end AI solution design, cloud architecture strategy, Agile delivery leadership, and executive stakeholder engagement. Targeting senior Solutions Architect and AI Platform leadership roles in the technology sectors.

CORE COMPETENCIES

AI/ML Solution Design	Cloud Architecture (Azure)	NLP & Conversational AI
Agile Delivery Leadership	Enterprise Data Integration	Stakeholder & Program Mgmt.
Python | SQL | Power BI	JIRA | Lucid chart | Visio	HIPAA | EHR | EDI

PROFESSIONAL EXPERIENCE

Senior AI Business Analyst & Solution Designer

September 2019 – Present

Gardner-White — MI

Key Initiative: Retail Operations Decision Assistant (Azure OpenAI + RAG)

Domain-specific GenAI solution grounding LLM responses in curated retail operations content to enable accurate, context-aware decision support for frontline staff.

- Designed secure, scalable AI architectures integrating Azure OpenAI with enterprise data systems, incorporating access control, data governance, and responsible AI practices to meet enterprise compliance and risk requirements.
- Designed and implemented a Retrieval-Augmented Generation (RAG) solution on Azure OpenAI and Azure AI Search, grounding LLM responses in a curated corpus of retail operations documents (return policies, delivery SOPs, warranty procedures) with hybrid vector and keyword retrieval, enabling frontline staff to surface accurate answers in seconds instead of searching across disconnected repositories.
- Optimized document chunking strategy and tuned hybrid retrieval parameters in Azure AI Search to improve answer relevance, reduce retrieval errors, and balance grounding quality against response latency.
- Developed a GenAI-powered document intelligence capability delivering automated summarization and context-aware insights across long-form business documents, with system prompts enforcing grounded-only responses and explicit "I don't know" fallbacks to prevent hallucination on out-of-scope questions.

- Engineered prompt strategies and tuned LLM response parameters (temperature, token limits, system instructions) for Azure OpenAI deployments, using rapid Playground-based iteration to improve response accuracy, reduce hallucinations, and produce consistent structured outputs that integrate cleanly with downstream enterprise systems.
- Designed system-level prompt frameworks to standardize AI behavior across enterprise applications, ensuring consistent, role-aware, and policy-aligned responses tailored to different user contexts and business rules.
- Established a lightweight evaluation framework using a labeled question set to measure retrieval accuracy and answer groundedness, iterating on top-k and chunk size parameters to improve RAG system performance before each release.
- Architected and deployed an enterprise conversational AI solution leveraging large language model (LLM) capabilities, enabling context-aware customer interactions and reducing support volume by 25%.
- Designed and optimized conversational AI workflows, integrating dynamic response generation with backend data systems to enable real-time, personalized customer interactions at scale.

Additional Responsibilities

- Partnered with product owners to define and prioritize a multi-quarter product roadmap, ensuring alignment between business strategies, customer needs, and technical feasibility across all delivery teams.
- Conducted gap analysis and SWOT assessments on existing e-commerce workflows, translating findings into actionable solution proposals presented to senior leadership for investment decisions.

Business Analyst — AI & Healthcare Systems

March 2017 – August 2019

Streamline Health Care — Michigan

US Healthcare / HIT

- Led the design of an enterprise AI virtual assistant leveraging NLP and knowledge-based reasoning, enabling automated clinical query resolution across symptom triage, appointment scheduling, medication inquiries, and billing navigation, and reducing administrative effort by 40%.
- Reduced human administrative effort by 40% by architecting HIPAA-compliant dynamic search and configuring the AI knowledge base for real-time clinical query resolution.
- Designed 60+ intents, 120+ entities, and complex medical-domain dialog flows tailored to clinical terminology, patient interaction patterns, and HIPAA regulatory compliance.
- Conducted 25+ stakeholder interviews with clinicians and compliance officers; authored the BRD and drove it through review, signoff, and sprint estimation.
- Developed and executed 100+ UAT test cases validating AI outputs against clinical accuracy benchmarks and HIPAA standards; collaborated with data science teams to improve model response quality.
- Engineered EHR data integration workflows, including patient identifier standards, ANSI X12/EDI compliance rules, and data matching algorithms to ensure clean records across all systems.

EDUCATION

Master of Science — Artificial Intelligence and Machine Learning

Indiana Wesleyan University | Concentration: Data Analytics

In Progress — 2026

Coursework: ML Fundamentals, Data Analytics, Neural Networks, AI Platform Design

Bachelor of Engineering — Electronics and Communication Engineering

Thanthai Periyar Government Institute of Technology, India | First Class with Honors

Electronics Engineering

ACADEMIC PROJECTS & PORTFOLIO

Netflix Data Analytics — End-to-End Data Pipeline & Visualization

Indiana Wesleyan University | DTAN-500 Foundations of Data Analytics

2024 – 2025

Cleaned and analyzed a large Netflix dataset using Excel and Python; built visualizations identifying content trends, genre distributions, and release patterns. Applied data fusion techniques to cross-validate findings and documented methodological decisions and dataset limitations.

Cricket Performance Analytics — Predictive Insights from Sports Data

Indiana Wesleyan University | DTAN-500 Foundations of Data Analytics

2024 – 2025

Performed statistical analysis on cricket match datasets using Excel and Python; compared manual analytics findings against ChatGPT-generated insights to evaluate AI-assisted analytics accuracy. Produced visualizations and a professional analytics report.

AI/ML Portfolio — Google Sites Professional Portfolio

Live Portfolio: sites.google.com/view/ramya-srinivasan-portfolio

In Progress

Curated portfolio showcasing data analytics projects, AI/ML coursework artifacts, and professional development work completed during the MS program at Indiana Wesleyan University.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Microsoft Azure Fundamentals (AZ-900)	Microsoft	*2025*
Microsoft Azure Administrator (AZ-104) In Progress	Microsoft	*Target: 2026*
Microsoft Azure Solutions Architect Expert (AZ-305) In Progress	Microsoft	*Target: 2026*
SAFe Certified Advanced Scrum Master	Scaled Agile	*Active*
Certified Business Analyst	Precision Technologies	*Active*

TECHNICAL SKILLS

AI / ML	Azure OpenAI, Retrieval-Augmented Generation (RAG), RAG Evaluation, Large Language Models (LLM), Prompt Engineering, GenAI Document Intelligence, Conversational AI, Responsible AI, Data Governance, NLP, Chatbot Architecture, ML Pipeline Design, Neural Networks, Python
Cloud	Microsoft Azure (AZ-900 certified; AZ-104 & AZ-305 in progress), Azure OpenAI Service, Azure AI Search, Azure AI Services, Cloud Architecture
Data & BI	Python, SQL, Power BI, Data Validation, ETL, Data Integration, EHR Systems, ANSI X12 / EDI
Delivery	Agile Leadership, SAFe, Kanban, JIRA Administration, Backlog Management, PI Planning
Architecture	Solution Design, Wireframing, Lucid chart, Visio, BRD Authoring, Gap & SWOT Analysis
Domains	US Healthcare (HIPAA, EHR, Clinical AI), Retail E-Commerce, AI Platform Integration, Compliance

KEY IMPACT METRICS

25%

Reduction in customer support queries via LLM-powered conversational AI

40%

Reduction in human effort through enterprise AI virtual assistant in healthcare

10+

Years of designing enterprise AI and analytics solutions across two industries

100+

UAT test cases designed and executed for AI/ML clinical output validation